

# High-chance portions - geo-physical

Combined original study articles for this topic's highest-probability PYQ themes.

## Monsoon & Climatology

~78% · Monsoon onset · Jet streams · ENSO · Climate classification · Rainfall patterns

### Why this portion (PYQ lens)

Portion chance ~78%. Steady across decade; parent geo-physical ~79%.

Years appeared (synthesis): 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025

### Why UPSC asks this

Mechanism statements and ENSO/monsoon links are classic Prelims geography.

### Syllabus framing

Portion chance ~78%. Steady across decade; parent geo-physical ~79%. This portion sits inside geo-physical and centres on Indian monsoon and climatology. Mechanism statements and ENSO/monsoon links are classic Prelims geography.

Recurring PYQ angles: Monsoon onset, Jet streams, ENSO, Climate classification, Rainfall patterns. Approximate appearance years in the 2016-2025 educational synthesis: 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025. Use the chance percentage only to sequence study hours - it is not an official prediction.

### Core map of Indian monsoon and climatology

Differential heating: Land - sea thermal contrast sets monsoon stage. ITC shift: Seasonal migration of wind belts influences inflow.

Somali jet / Findlater: Cross-equatorial flow into Arabian Sea branch narrative. Southwest monsoon branches: Arabian Sea and Bay of Bengal branches.

Hold this core map as your elimination spine before memorising peripheral CA overlays.

### Core concepts and definitions

Orography: Western Ghats and Himalaya shape rainfall distribution. Break and active periods: Intra-seasonal monsoon variability.

Northeast monsoon: Tamil Nadu rainfall season contrast. ENSO: El Niño often linked to weaker monsoon - probabilistic not absolute.

### Institutions, processes and mechanisms

IOD: Indian Ocean Dipole modulates monsoon independently at times. Western Disturbances: Winter/spring precip in northwest India.

Jet streams: Subtropical jet / easterly jet roles in onset narratives. Climate classification: Koeppen-type codes for India - revise map.

### India angle and applied examples

Climate change overlay: Monsoon statistics shifting - careful wording. Monsoon trough: Location influences rainfall belts.

Bay lows/depressions: Rain producers in peak season. Retreating monsoon: September - December transitions.

## Comparisons, linkages and discriminators

Rain-shadow: Leeward dryness - Ghats example. Mango showers / blossom showers: Local pre-monsoon terms.

Onset criteria: IMD operational definitions - conceptual. Flood vs drought years: Teleconnections + local factors.

## India angle

IMD onset/progress maps are CA; mechanism remains static spine.

## Question patterns and exam spine

Matching lists and statements on Indian monsoon and climatology.

Revision spine: Definitions -> table -> India -> traps.

Eliminate anachronisms and category errors first, then test remaining options against table rows and must-remember triggers.

## Trap watch-list (specific)

Avoid: El Niño always means full monsoon failure. Avoid: Ignoring orography. Avoid: Mixing NE and SW monsoon regions. Avoid: Absolute jet-stream claims.

## Monsoon quick map

### Idea | Hook

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SW monsoon -- June - Sep main

NE monsoon -- TN focus

ENSO -- Probabilistic link

Orography -- Ghats/Himalaya

## Map / list facts

- Western Ghats rain-shadow
- Bay of Bengal branch path
- TN NE monsoon

## Common traps

- x El Niño always means full monsoon failure
- x Ignoring orography
- x Mixing NE and SW monsoon regions
- x Absolute jet-stream claims

## Must remember

- Two SW branches
- Orography role
- ENSO probabilistic
- NE monsoon TN
- Jet/onset narrative
- Koeppen India map

## Quick revision checklist

- ☐ Reproduce table
- ☐ 5 must-remembers
- ☐ 3 traps
- ☐ India anchor
- ☐ 10 MCQs

Educational prioritisation heuristic - not an official UPSC prediction. (c) UPSC IAS Study Hub original notes.

# Geomorphology & Oceanography

~72% · Plate tectonics · Earthquakes · Landforms · Ocean currents · Coral reefs

## Why this portion (PYQ lens)

Portion chance ~72%. Recurring; parent geo-physical ~79%.

Years appeared (synthesis): 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025

## Why UPSC asks this

Process - landform matching and current maps are durable PYQ styles.

## Syllabus framing

Portion chance ~72%. Recurring; parent geo-physical ~79%. This portion sits inside geo-physical and centres on geomorphology and oceanography. Process - landform matching and current maps are durable PYQ styles.

Recurring PYQ angles: Plate tectonics, Earthquakes, Landforms, Ocean currents, Coral reefs. Approximate appearance years in the 2016-2025 educational synthesis: 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025. Use the chance percentage only to sequence study hours - it is not an official prediction.

## Core map of geomorphology and oceanography

Plate boundaries: Convergent/divergent/transform - earthquake/volcano logic. Himalayan orogeny: Collision continuum - seismic implications.

Weathering vs erosion: In-situ decay vs transport removal. Fluvial landforms: Meanders, oxbow, deltas, terraces.

Hold this core map as your elimination spine before memorising peripheral CA overlays.

## Core concepts and definitions

Glacial landforms: Cirque, moraine - limited India but asked. Coastal landforms: Bars, lagoons, cliffs, mangroves interface.

Karst: Limestone solution landscapes. Earthquake focus/epicentre: Definitions.

## Institutions, processes and mechanisms

Tsunami generation: Displacement of water column. Volcanic types: Shield vs composite outline.

Ocean currents: Warm/cold; climate influence on coasts. Upwelling: Nutrient-rich fisheries link.

## India angle and applied examples

Tides: Gravitational; spring/neap idea. EEZ vs territorial sea: UNCLOS vocabulary bridge.

Coral reefs: Conditions: warm, clear, shallow - bleaching CA. Continental shelf: Resource and hazard relevance.

## Comparisons, linkages and discriminators

Cyclone ocean heat: Warm SST energy source. Seamounts/trenches: Map terms.

Mass movements: Landslides in Himalaya - DM bridge. Isostasy outline: Vertical adjustment idea.

## India angle

Himalayan seismicity + Indian Ocean tsunami literacy + coral bleaching headlines.

## Question patterns and exam spine

Matching lists and statements on geomorphology and oceanography.

Revision spine: Definitions -> table -> India -> traps.

Eliminate anachronisms and category errors first, then test remaining options against table rows and must-remember triggers.

## Trap watch-list (specific)

Avoid: All plate edges identical hazards. Avoid: Cold currents always dry coasts absolutely. Avoid: Coral in cold deep seas. Avoid: Focus=epicentre.

## Boundary hazards

### Boundary | Typical association

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Convergent -- Quakes/volcanoes/orogeny

Divergent -- Rifting volcanism

Transform -- Strike-slip quakes

## Map / list facts

- Ring of Fire contrast
- Indian Ocean currents sketch
- Andaman trench awareness

## Common traps

- x All plate edges identical hazards
- x Cold currents always dry coasts absolutely
- x Coral in cold deep seas
- x Focus=epicentre

## Must remember

- 3 boundary types
- Weathering != erosion
- Warm vs cold currents idea
- Coral conditions
- Focus vs epicentre

## Quick revision checklist

- [ ] Reproduce table
- [ ] 5 must-remembers
- [ ] 3 traps
- [ ] India anchor
- [ ] 10 MCQs

Educational prioritisation heuristic - not an official UPSC prediction. (c) UPSC IAS Study Hub original notes.